

JC2002 Java Programming - Practical 1

The aim of this practical is to get familiar with Java Integrated Development Environment (IDE) - we recommend IntelliJ IDEA Community edition in this course - and create a GitHub student account (if you do not have one already), which will be useful during your university studies.

1. You should download a Java IDE for this course to run your applications, and to program them too (Keep in mind – you have to submit a programming coursework after this course! So it will be never too late to start practicing). We do recommend IntelliJ IDEA Community edition, which is the free, open-source version of JetBrains' powerful Integrated Development Environment (IDE) specifically designed for Java and a few other JVM languages. It's built on the same core platform as the paid Ultimate edition but with a more focused feature set.
2. GitHub is a cloud repository for source code both public and private. Git is a version control system to keep track of your applications as they grow, and GitHub is an online repository to store them for individuals, organisations, and companies. You should [create a student account](#) now to use all through your university studies. This will allow you private (instead public) repositories, which you should always use for your assessment work.

Use GitHub regularly to store all of your assessment work to avoid the hassles of when your laptop fails you at a crucial time. You are a computing student, so now is the time to start your good backup habits.

Do the 'hello world' tutorial at GitHub.

3. Next, you should get familiar with editing source code, as well as compiling and running Java programs in IntelliJ IDEA. For your convenience, you can split the panel view in file editing area and terminal area.

Create a new Java source file. You should name the file as 'HelloWorld.java'. In the file, you should write the Java code given below (you can paste it from this document, or write it manually, the code is not very long):

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello world!");  
    }  
}
```

Remember to save the file after you made any changes. When the file is ready, you can try to compile by writing the following command in the terminal window:

```
javac HelloWorld.java
```

If the file is compiled successfully, file HelloWorld.class should appear in the directory. Then, you can execute the file by writing command:

```
java HelloWorld
```

If everything went correctly, text “Hello World” should appear in the terminal window. You can try to modify the text “Hello World” in the source code to something else and then try to recompile and run the program again.